

Revision Notes for Class 12 Maths

Chapter 1 – Relations and Functions

Relation

- Relations in Maths is one of the very important topics for the set theory.
- Relations and functions generally tell us about the different operations performed on the sets.
- Relation in Maths can be put into term as a connection between the elements of two or more sets and the sets must be non-empty.
- A relation namely R is formed by a Cartesian product of subsets.
- It defines the relationship between two sets of values, let say from set A to set B .
- Set A is then called domain and set B is then called codomain. If $(a,b) \in R$, it shows that a is related to b under the relation R

Types of Relations:

1. Empty Relation:

- In this there is no relation between any element of a set.
- It is also known as void relation
- For example: if set A is $\{2,4,6\}$ then an empty relation can be $R = \{x,y\}$ where $x+y > 11$

2. Universal Relation:

- In this each element of a set is related to every element of that set.
- For example: if set A is $\{2, 4, 6\}$ then a universal relation can be $R = \{x, y\}$ where $x + y > 0$

3. Trivial Relation: Empty relation and universal relation is sometimes called trivial relation.

4. Reflexive Relation:

- In this each element of set (say) A is related to itself i.e., a relation R in set A is called reflexive if $(a, a) \in R$ for every $a \in A$.
- For example: if $Set A = \{1, 2, 3\}$ then relation $R = \{(1, 1), (1, 2), (2, 2), (2, 1), (3, 3)\}$ is reflexive since each element of set A is related to itself.

5. Symmetric Relation:

- A relation R in set A is called symmetric if $(a, b) \in R$ and $(b, a) \in R$ for every $a, b \in A$.
- For example: if $Set A = \{1, 2, 3\}$ then relation $R = \{(1, 2), (2, 1), (2, 3), (3, 2), (3, 1), (1, 3)\}$ is symmetric.

6. Transitive Relation:

- A relation R in set A is called transitive if $(a, b) \in R$ and $(b, c) \in R$ then (a, c) also belongs to R for every $a, b, c \in A$.
- For example: if $Set A = \{1, 2, 3\}$ then relation $R = \{(1, 2), (2, 3), (1, 3), (2, 3), (3, 2), (2, 2)\}$ is transitive.

7. Equivalence Relation:

- A relation R on a set A is equivalent if R is reflexive, symmetric and transitive.
- For example: $R = \{(L_1, L_2) : \text{line } L_1 \text{ is parallel to line } L_2\}$,

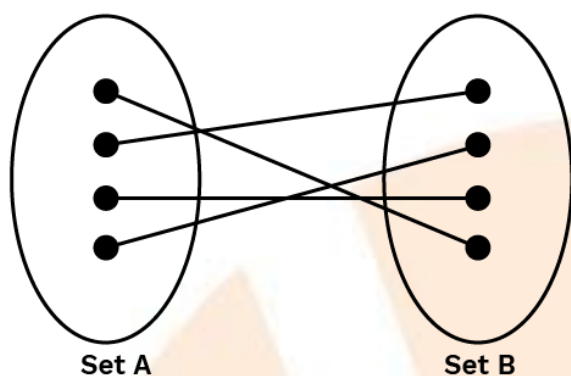
This relation is reflexive because every line is parallel to itself

Symmetric because if L_1 parallel to L_2 then L_2 is also parallel to L_1

Transitive because if L_1 parallel to L_2 and L_2 parallel to L_3 then L_1 is also parallel to L_3

Functions

A function can have the same range mapped as that of in relation, such that a set of inputs is related to exactly one output. A function f from a set A to a set B is a rule which associates each element of set A to a unique element of set B .



- Set A is domain and set B is codomain of the function
- Range is the set of all possible resulting values given by the function.

For example: x^2 is a function where values of x will be the domain and value given by x^2 is the range.

Types of Function:

1. One-One Function:

- A function f from set A to set B is called one-one function if no two distinct elements of A have the same image in B.
- Mathematically, a function f from set A to set B if $f(x) = f(y)$ implies that $x = y$ for all $x, y \in A$.
- One-one function is also called an injective function.
- For example: If a function f from a set of real numbers to a set of real numbers, then $f(x) = 2x$ is one-one function.

2. Onto Function:

- A function f from set A to set B is called onto function if each element of set B has a preimage in set A or range of function f is equal to the codomain i.e., set B .
- Onto function is also called surjective function.
- For example: If a function f from a set of natural numbers to a set of natural numbers, then $f(x) = x - 1$ is onto the function.

3. Bijective Function:

- A function f from set A to set B is called a bijective function if it is both one-one function and onto function.
- For example: If a function f from a set of real numbers to a set of real numbers, then $f(x) = 2x$ is one-one function and onto function.

Composition of function and invertible function

- **Composition of function:** Let $f : A \rightarrow B$ and $g : B \rightarrow C$ then the composite of g and f , written as $g \circ f$ is a function from A to C such that $(g \circ f)(a) = g(f(a))$ for all $a \in A$. (Not in the current syllabus)
- **Properties of composition of function:** Let $f : A \rightarrow B$, $g : B \rightarrow C$ and $h : C \rightarrow A$ then

a. Composition is associative i.e., $h(gf) = (hg)f$

b. If f and g are one-one then $g \circ f$ is also one-one

c. If f and g are onto then $g \circ f$ is also onto

d. Invertible function: If f is bijective then there is a function $f^{-1}:B \rightarrow A$ such that $(f^{-1}f)(a)=a$ for all $a \in A$ and $(ff^{-1})(b)=b$ for all $b \in B$

f^{-1} is the inverse of the function f and is always unique. (Not in the current syllabus)

Binary Operations

- A binary operation are mathematical operations such as addition, subtraction, multiplication and division performed between two operands.
- A binary operation on a set A is defined as operations performed between two elements of set A and the result also belongs to set A . Then set A is called binary composition.
- It is denoted by $*$

For example: Binary addition of real numbers is a binary composition since by adding two real numbers the result will always be a real number.

Properties of Binary Composition:

- A binary operation $*$ on the set X is commutative, i.e., $a*b=b*a$, for every $a,b \in X$
- A binary operation $*$ on the set X is associative, i.e., $a*(b*c)=(a*b)*c$, for every $a,b,c \in X$
- There exists identity for the binary operation $*:A \times A \rightarrow A$, i.e., $a*e=e*a=a$ for all $a,e \in A$

A binary operation $*:A \times A \rightarrow A$ is said to be invertible with respect to the operation $*$ if there exist an element b in A such that $a*b=b*a=e$, e is identity element in A then b is the inverse of a and is denoted by a^{-1}

Relations and Functions Class 12 Notes Mathematics

All the topics and subtopics which are covered in Relations and Functions for Class 12 are given below:

- Introduction
- Types of Relations
- Types of Functions
- Composition of functions and invertible functions
- Binary operations